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Final Project Report 1

CNN Architectures, Regularization, and Data Augmentation for RGB and Infrared ADAS Object Recognition

Course	CSE 547 - Deep Learning Algorithms and Methods
Section	CSE 547_51 | Spring 2026
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Abstract. This report evaluates RGB and infrared object recognition on the FLIR ADAS v2 dataset for eight autonomous-driving classes: bike, bus, car, person, sign, motor, light, and truck. Part 1 compares four CNN architectures per modality with approximately 10x parameter growth between consecutive models. Part 2 selects a modestly overfitting model per modality and evaluates L2 regularization, dropout, data augmentation, and combined regularization settings. The best raw Part 1 model was Arch D for both modalities, with RGB reaching Val F1 = 0.9185 and IR reaching Val F1 = 0.8792. In Part 2, IR benefited most from moderate data augmentation, reaching Val F1 = 0.9227, while RGB benefited most from light dropout on Arch C, reaching Val F1 = 0.9099.

1. Dataset and Experimental Setup

The experiments use extracted object patches from the FLIR ADAS v2 dataset. The IR manifest contains 52,930 thermal patches and the RGB manifest contains 108,297 RGB patches. Each patch is labeled as one of eight classes. To prevent leakage from nearby frames and repeated scene conditions, training and validation splits were assigned at the video level for each modality.

All patch classifiers used 64 x 64 RGB inputs, Adam optimization with learning rate 1e-3, cross-entropy loss, batch size 256, and weighted F1 as the primary validation metric. Training was configured for at least 50 epochs with early stopping patience of 10 epochs, but early stopping was only eligible after epoch 50. The maximum epoch count was 70.

Table 1. Patch distribution by modality.

Class	IR patches	RGB patches
car	31,513	54,462
person	12,432	25,030
bike	3,105	6,304
sign	2,021	11,358
bus	1,721	1,629
light	848	7,075
motor	730	1,408
truck	560	1,031
Total	52,930	108,297

2. Part 1 - CNN Architecture Comparison

Four CNN architectures were trained per modality. The parameter counts were 880, 14,600, 148,200, and 2,459,848, satisfying the required 10x growth rule between consecutive architectures.

Table 2. CNN architecture summary.

Model	Architecture summary	Parameters
Arch A	2 conv layers with 8 filters, global average pooling, linear output	880
Arch B	Conv 16, Conv 32, Conv 32, global average pooling, linear output	14,600
Arch C	2x Conv 32, 2x Conv 64, Conv 128, global average pooling, dense head	148,200
Arch D	2x Conv 64, 2x Conv 128, 2x Conv 256, Conv 512, batch norm, dense head	2,459,848

Table 3. Part 1 IR architecture results.

Model	Params	Best Ep	Train Acc	Val Acc	Train F1	Val F1
Arch A	880	41	0.7663	0.7251	0.7168	0.6520
Arch B	14,600	62	0.8702	0.8070	0.8507	0.7563
Arch C	148,200	64	0.9552	0.8616	0.9535	0.8340
Arch D	2,459,848	9	0.9766	0.8880	0.9772	0.8792

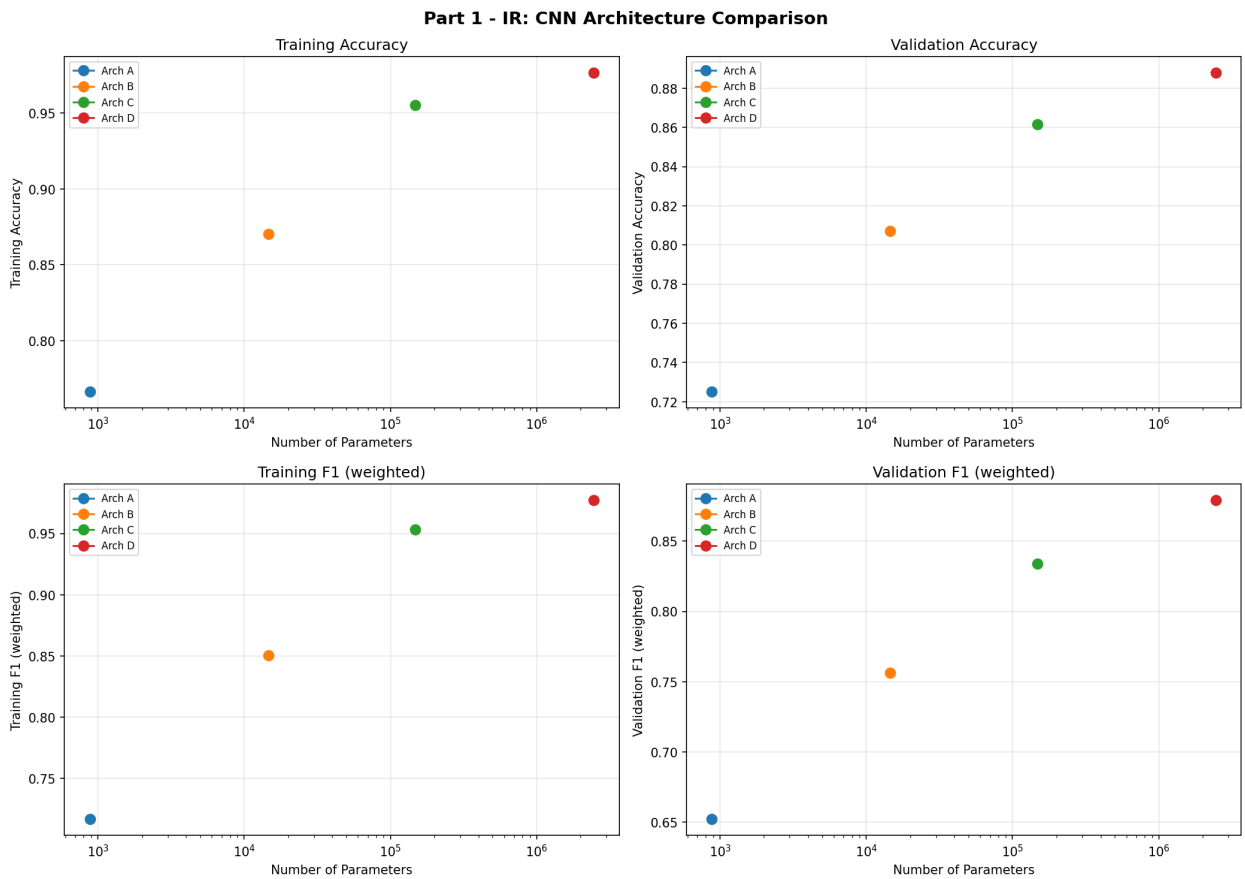

Figure 1. IR architecture comparison. Validation performance improves as model capacity increases, with Arch D producing the best IR weighted F1.

For IR, model capacity was strongly associated with validation performance. Arch A underfit, while Arch B and Arch C improved both accuracy and weighted F1. Arch D achieved the best IR validation result with Val Acc = 0.8880 and Val F1 = 0.8792, although it also showed a clear train-validation gap.

Table 4. Part 1 RGB architecture results.

Model	Params	Best Ep	Train Acc	Val Acc	Train F1	Val F1
Arch A	880	24	0.6249	0.5988	0.5785	0.5384
Arch B	14,600	64	0.8393	0.7959	0.8230	0.7665
Arch C	148,200	70	0.9468	0.8909	0.9453	0.8857
Arch D	2,459,848	39	0.9999	0.9201	0.9999	0.9185

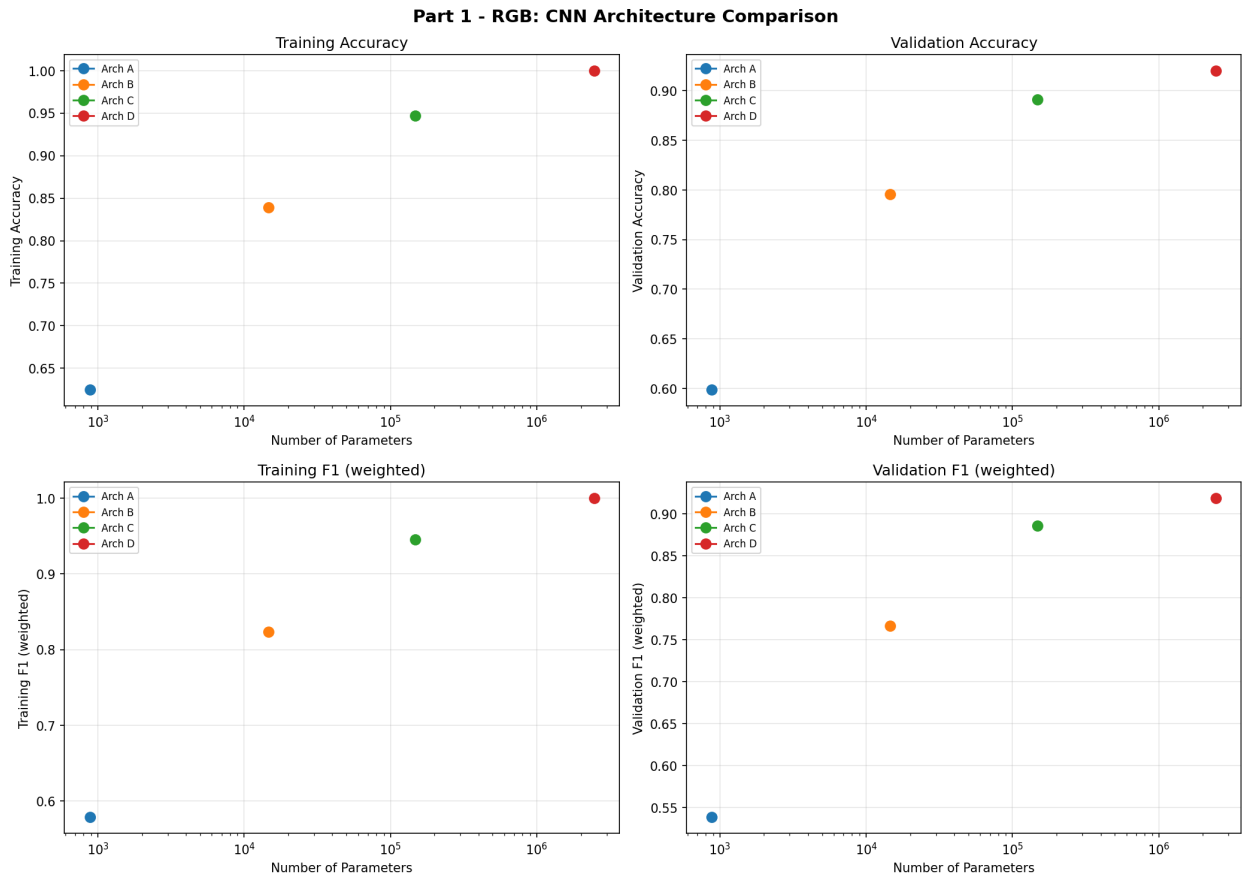

Figure 2. RGB architecture comparison. The largest architecture, Arch D, achieved the strongest RGB validation accuracy and weighted F1.

For RGB, the same trend was observed: increasing model capacity improved validation performance. Arch D achieved the strongest Part 1 RGB result with Val Acc = 0.9201 and Val F1 = 0.9185. RGB therefore outperformed IR in the best raw architecture comparison. However, Arch D showed a larger train-validation gap than Arch C, so Arch C was selected for the RGB Part 2 regularization experiments as the more defensible modest-overfitting base model.

3. Part 2 - IR Regularization and Augmentation

For IR, Arch D was selected from Part 1 because it produced the best validation performance and had sufficient capacity for meaningful regularization experiments. Four groups of experiments were performed: L2 regularization, dropout, data augmentation, and combined settings.

Table 5. Data augmentation level definitions.

Level	Transformations
Aug L1	Random horizontal flip only
Aug L2	Flip, rotation 10 degrees, affine translation 0.05
Aug L3	Flip, rotation 20 degrees, affine translation 0.10, color jitter 0.2
Aug L4	Flip, rotation 30 degrees, affine translation 0.15, color jitter 0.3

Table 6. IR L2 regularization results.

Setting	Val Acc	Val F1
L2 = 1e-5	0.8984	0.8984
L2 = 1e-4	0.8862	0.8903
L2 = 1e-3	0.8724	0.8491
L2 = 1e-2	0.9064	0.8973

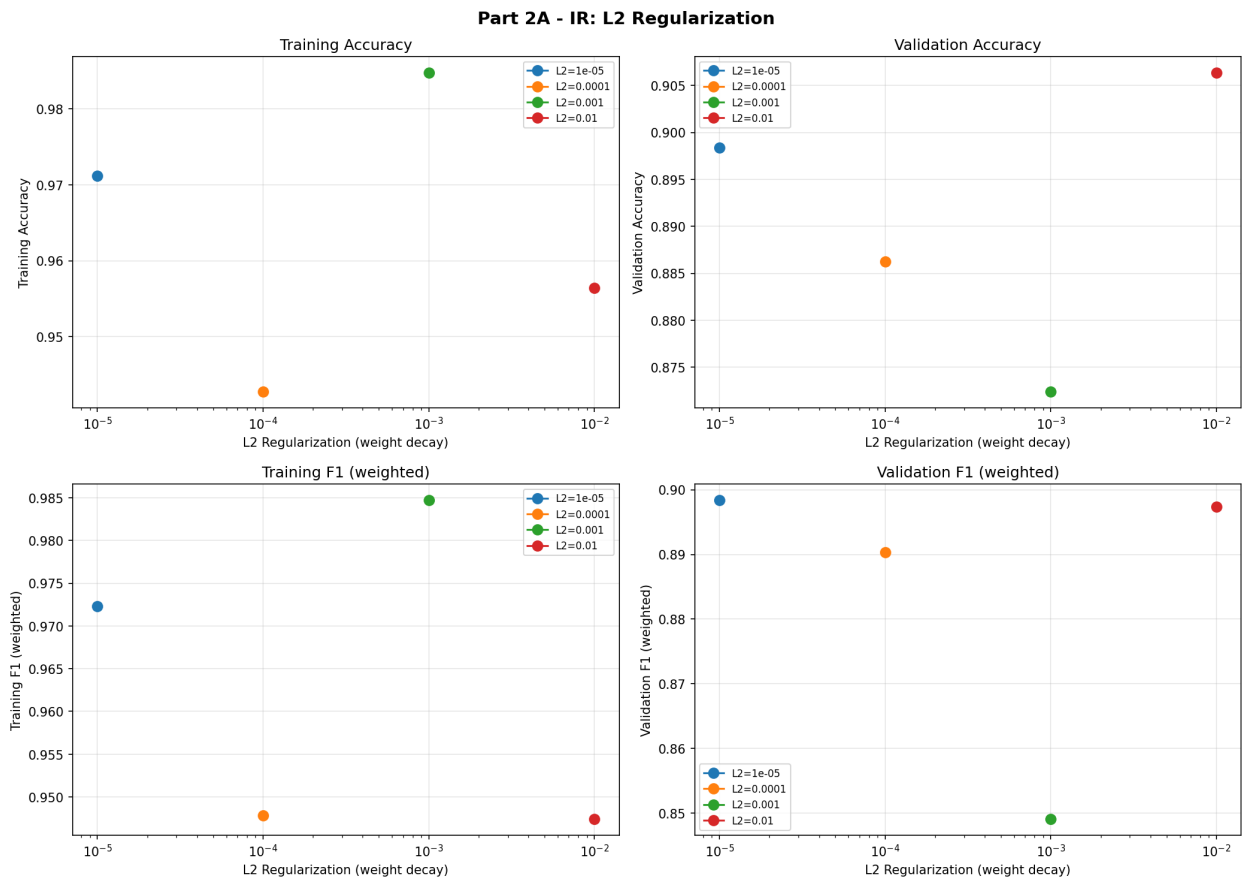

Figure 3. IR L2 regularization sweep.

Table 7. IR dropout results.

Setting	Val Acc	Val F1
Dropout = 0.10	0.8984	0.8838
Dropout = 0.25	0.8924	0.8743
Dropout = 0.40	0.8977	0.8809

Setting	Val Acc	Val F1
Dropout = 0.60	0.8746	0.8420

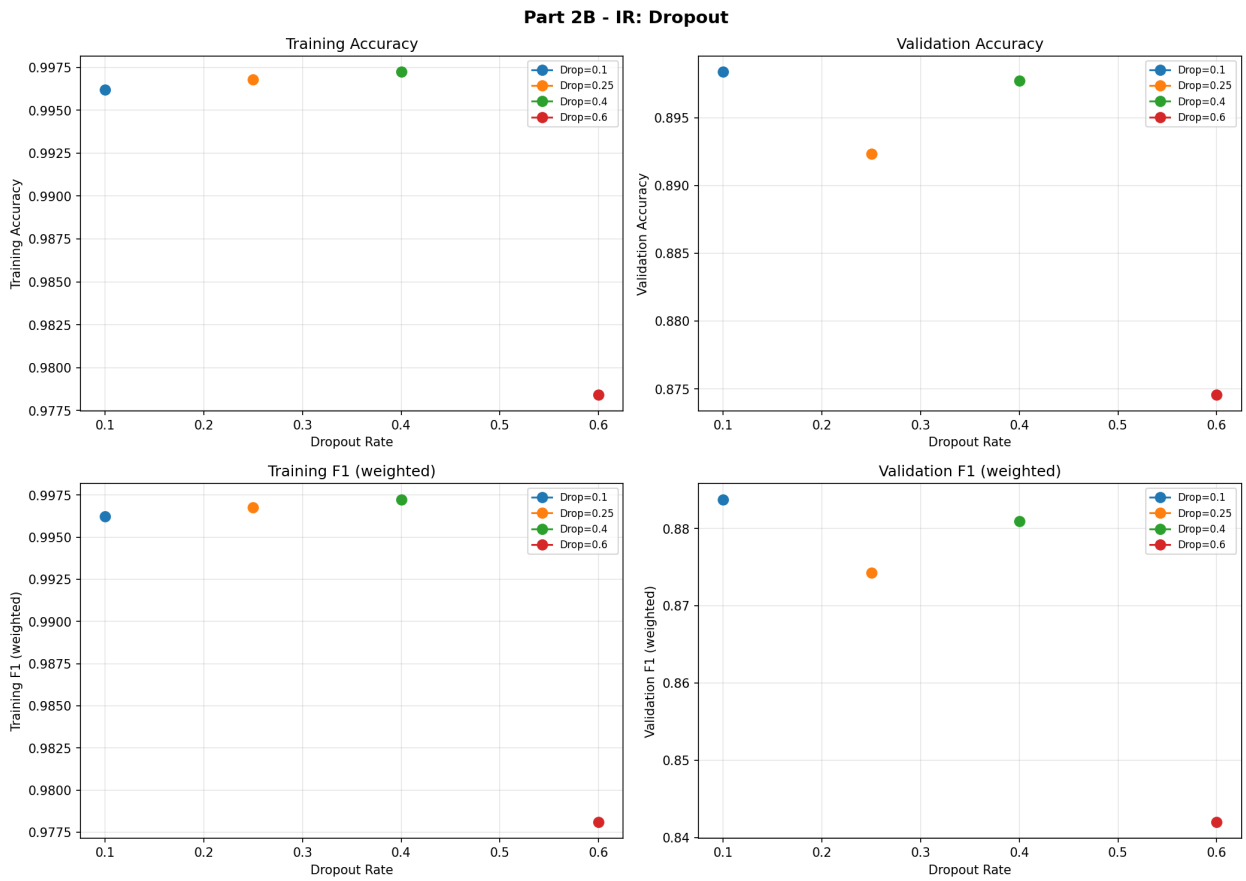

Figure 4. IR dropout sweep.

Table 8. IR data augmentation results.

Setting	Val Acc	Val F1
Aug L1	0.9085	0.8967
Aug L2	0.9263	0.9227
Aug L3	0.8908	0.8837
Aug L4	0.8865	0.8872

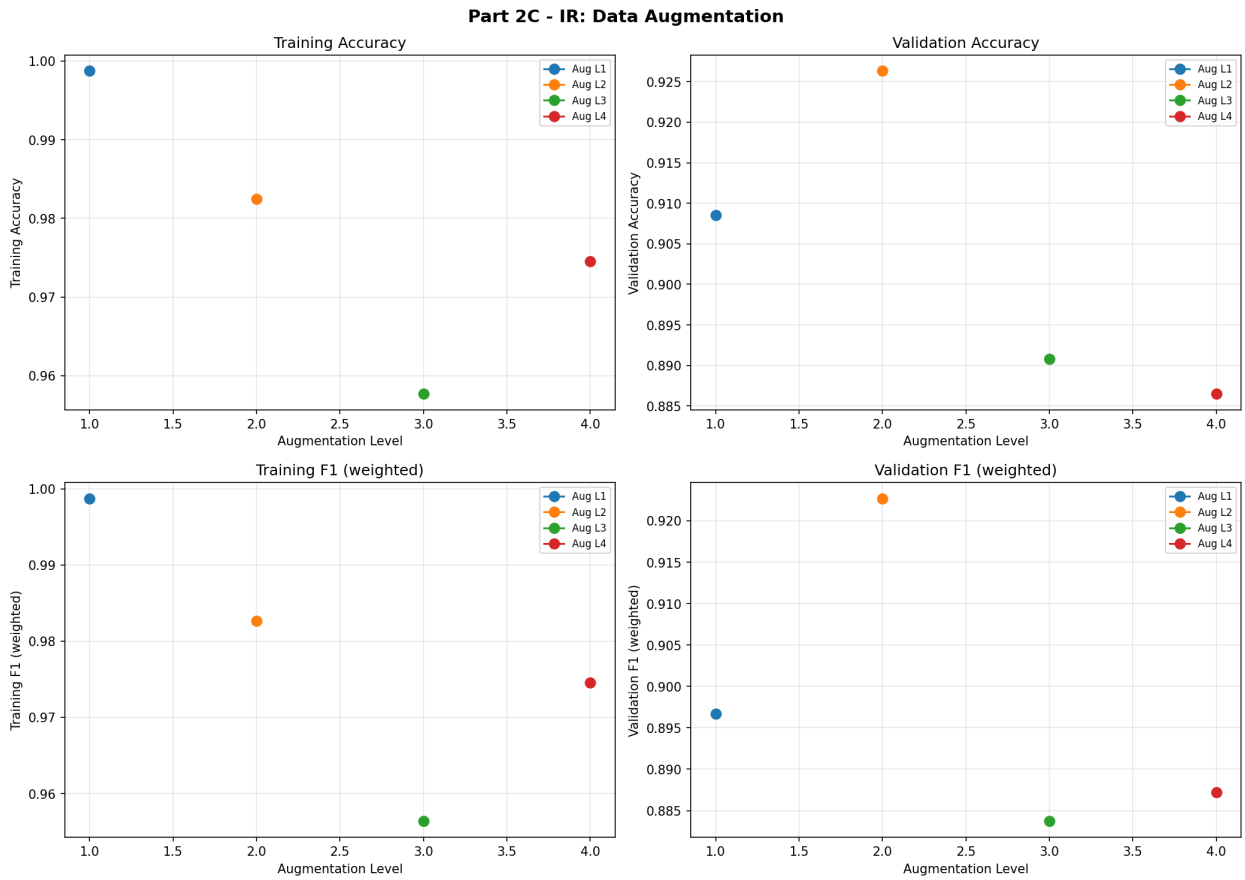

Figure 5. IR data augmentation sweep. Augmentation level 2 produced the strongest IR result.

Table 9. IR combined regularization results.

Setting	Val Acc	Val F1
L2 = 1e-5 + Dropout = 0.1	0.8786	0.8601
L2 = 1e-5 + Aug L2	0.9129	0.9035
Dropout = 0.1 + Aug L2	0.8988	0.8837
L2 + Dropout + Aug	0.9057	0.8981

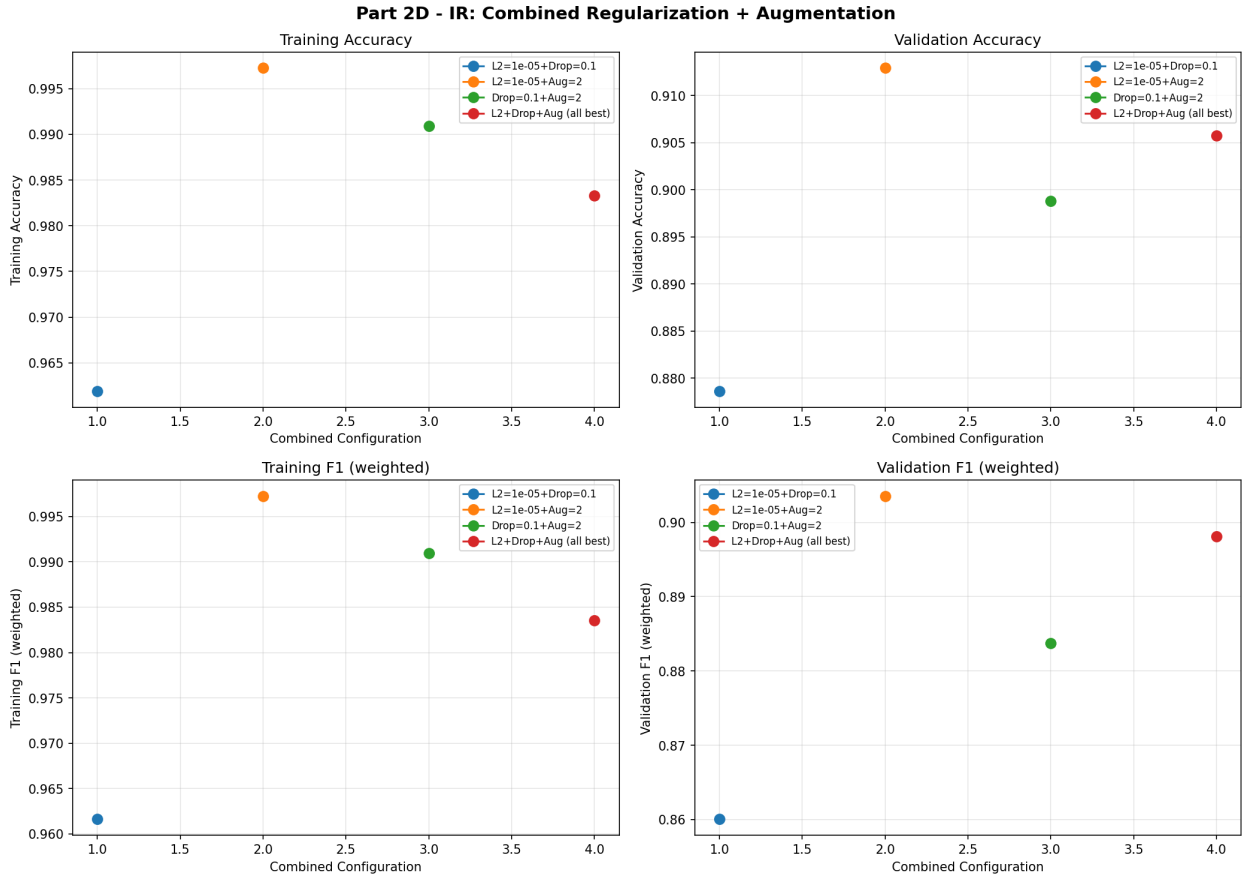

Figure 6. IR combined regularization sweep.

The strongest IR Part 2 result was Aug L2 with Val Acc = 0.9263 and Val F1 = 0.9227. This improved over the Part 1 IR Arch D baseline by 0.0435 F1. The combined settings did not exceed the best augmentation-only model, suggesting that regularization effects were not additive for this modality.

4. Part 2 - RGB Regularization and Augmentation

For RGB, Arch C was selected as the base model for Part 2 because it exhibited moderate overfitting while maintaining strong validation performance. This avoided using Arch D, which achieved the best raw Part 1 score but nearly memorized the training set.

Table 10. RGB L2 regularization results.

Setting	Val Acc	Val F1
L2 = 1e-5	0.8883	0.8835
L2 = 1e-4	0.8965	0.8894
L2 = 1e-3	0.8495	0.8335
L2 = 1e-2	0.6777	0.5766

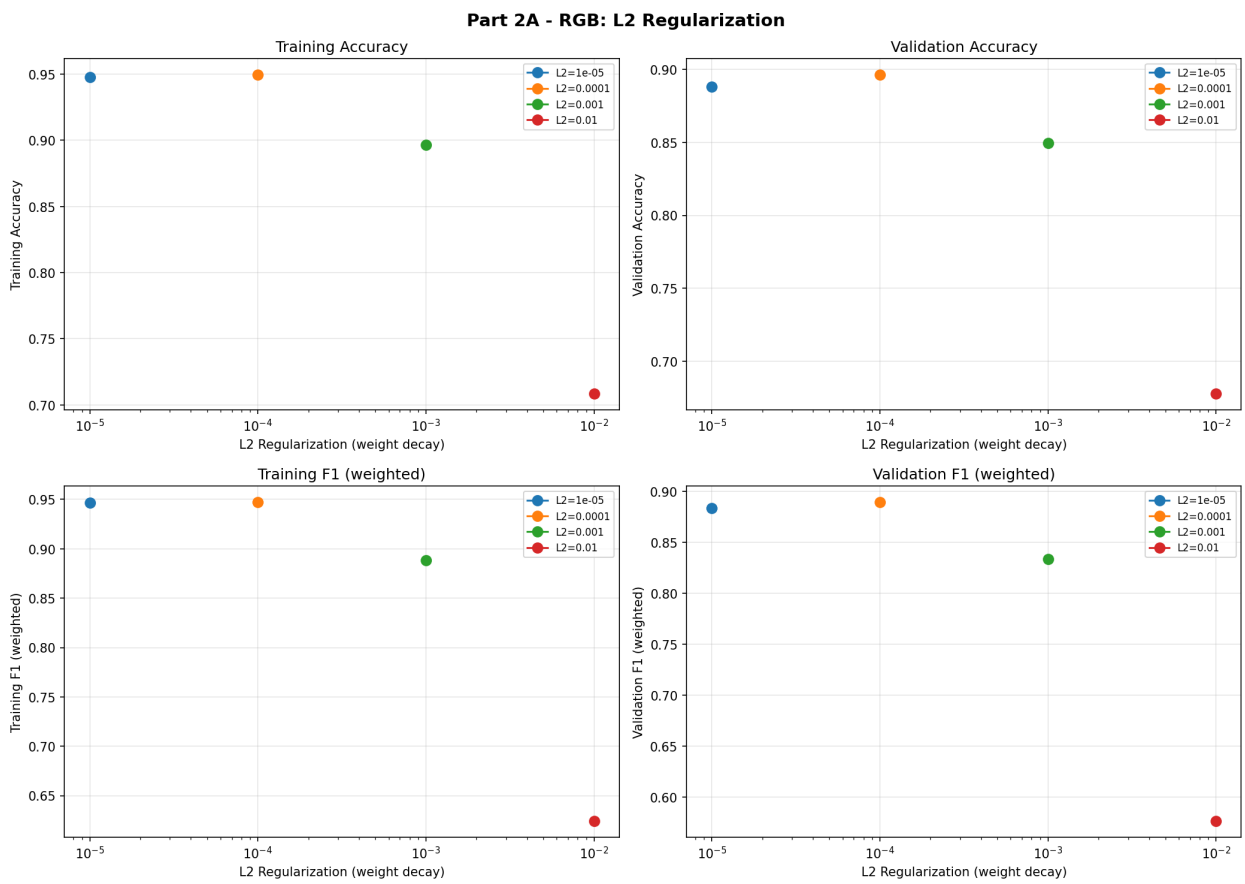

Figure 7. RGB L2 regularization sweep.

Table 11. RGB dropout results.

Setting	Val Acc	Val F1
Dropout = 0.10	0.9144	0.9099
Dropout = 0.25	0.8898	0.8778
Dropout = 0.40	0.8801	0.8602
Dropout = 0.60	0.8453	0.8162

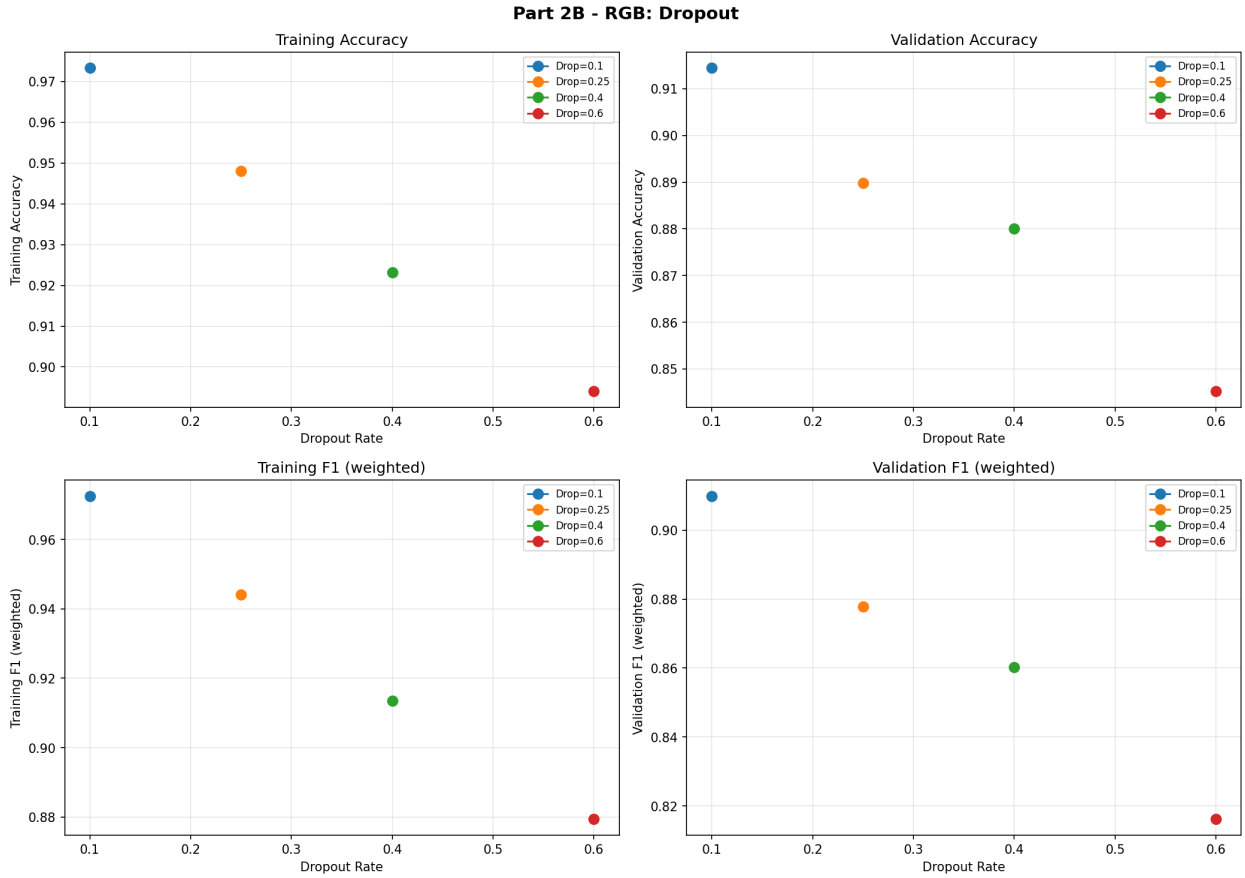

Figure 8. RGB dropout sweep. Dropout = 0.1 produced the best RGB Part 2 result.

Table 12. RGB data augmentation results.

Setting	Val Acc	Val F1
Aug L1	0.9094	0.9059
Aug L2	0.9043	0.8997
Aug L3	0.8846	0.8765
Aug L4	0.8447	0.8231

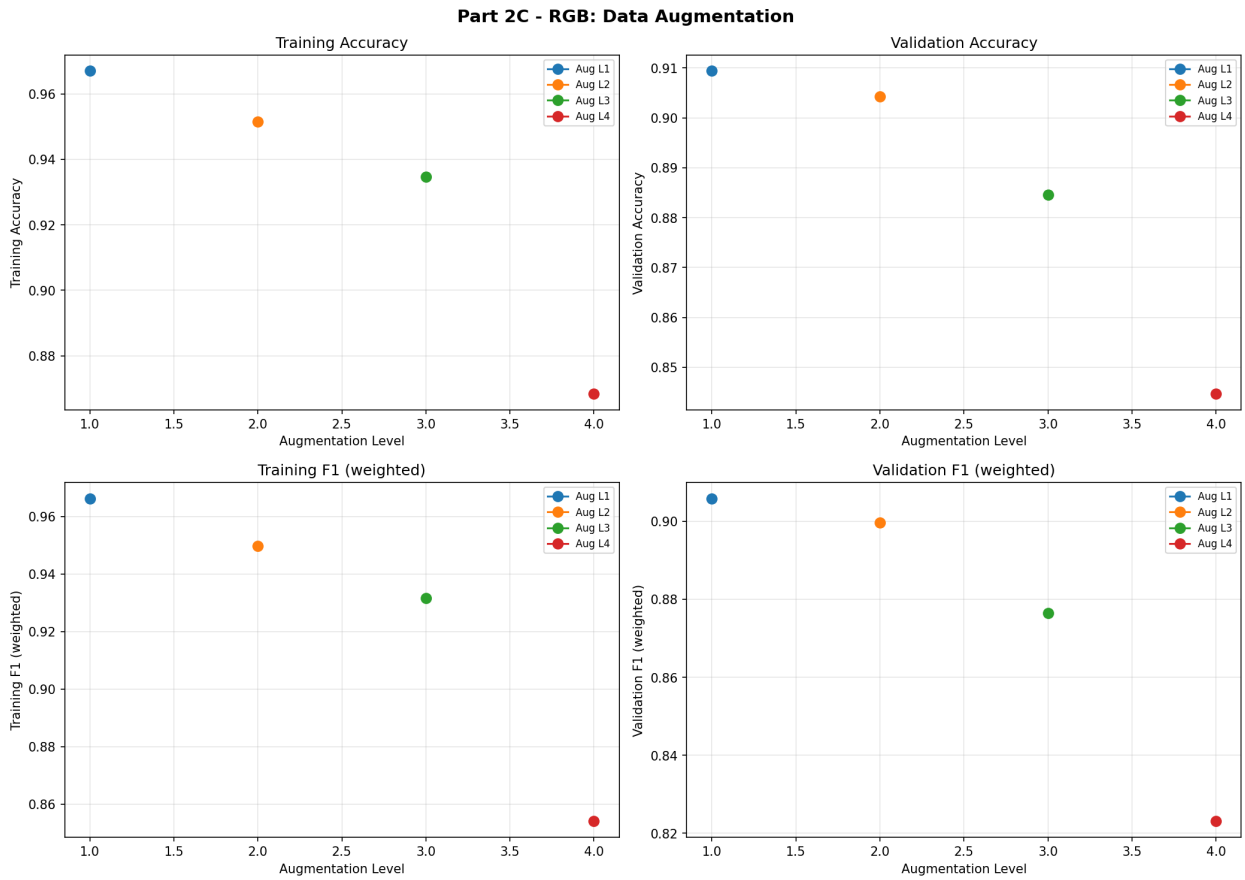

Figure 9. RGB data augmentation sweep.

Table 13. RGB combined regularization results.

Setting	Val Acc	Val F1
L2 = 1e-4 + Dropout = 0.1	0.8994	0.8947
L2 = 1e-4 + Aug L1	0.8818	0.8721
Dropout = 0.1 + Aug L1	0.9091	0.9032
L2 + Dropout + Aug	0.8909	0.8744

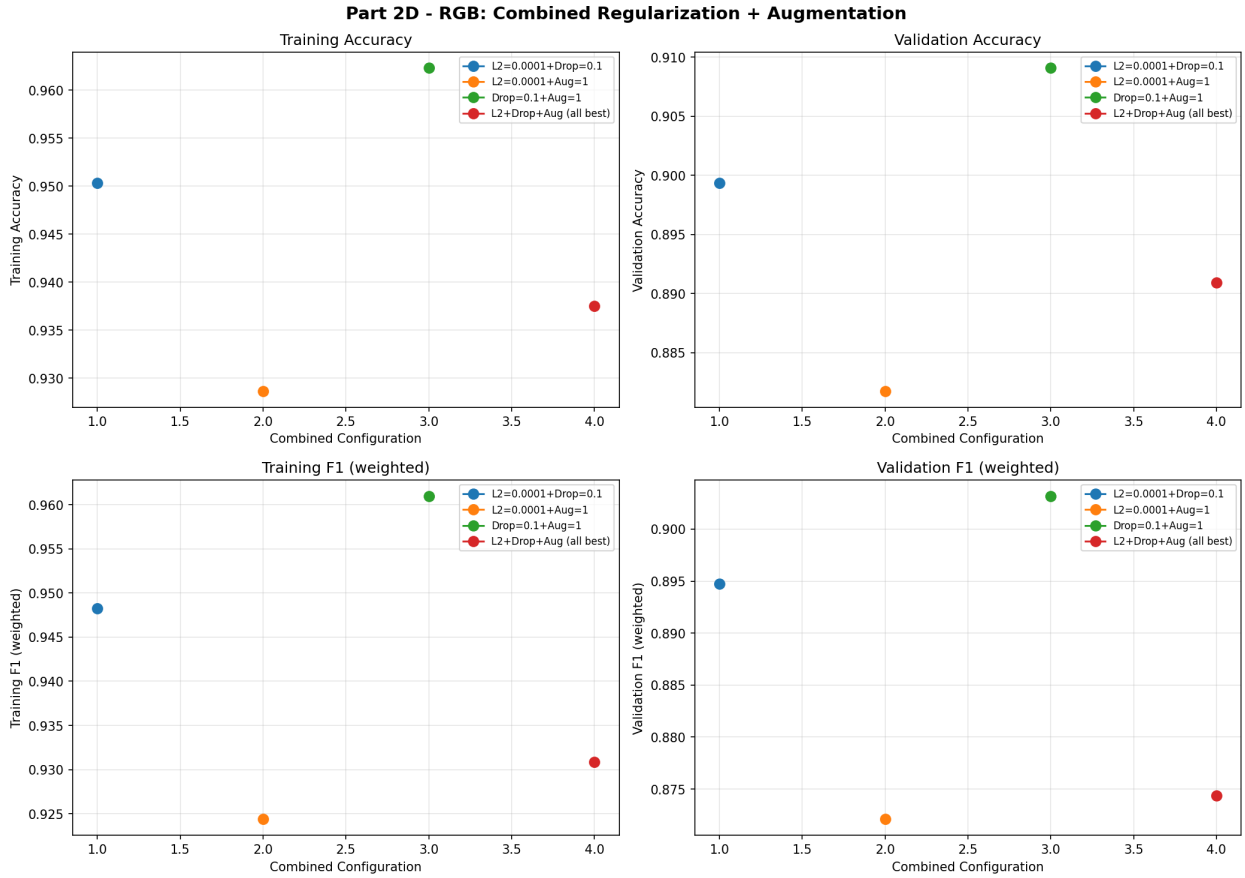

Figure 10. RGB combined regularization sweep.

The best RGB Part 2 result was dropout = 0.1 with Val Acc = 0.9144 and Val F1 = 0.9099. This improved over the Part 1 Arch C baseline by 0.0242 F1. Mild augmentation also helped, but stronger geometric and color transformations reduced performance. As with IR, combining the individually best options did not outperform the best single regularization strategy.

5. Cross-Modality Discussion

The Part 1 results show that RGB achieved higher raw validation performance than IR when both modalities used their best CNN architecture. This likely reflects the richer texture, color, and edge detail available in RGB object patches. However, IR responded more strongly to data augmentation in Part 2: moderate augmentation raised IR Val F1 from 0.8792 to 0.9227, exceeding the best RGB Part 2 result on the selected Arch C base.

The strongest regularization strategy differed by modality. IR benefited most from moderate geometric augmentation, which likely helped the model become less sensitive to object position and thermal-shape variation. RGB benefited most from light dropout, while aggressive augmentation reduced validation performance. This suggests that RGB patches already contain discriminative visual details that can be damaged by strong transformations, whereas IR patches benefit from moderate robustness training.

Another important observation is that combining the best regularization settings did not necessarily produce the best model. In both modalities, the best combined option underperformed the best single option. This indicates interaction effects between regularization methods and supports evaluating each method empirically rather than assuming additive improvements.

6. Conclusions

- Increasing CNN capacity improved validation performance in both RGB and IR experiments, and the required 10x parameter growth rule was satisfied.

- Arch D was the best Part 1 architecture for both modalities, reaching Val F1 = 0.8792 for IR and Val F1 = 0.9185 for RGB.

- For IR Part 2, augmentation level 2 was the best overall regularization strategy, reaching Val F1 = 0.9227.

- For RGB Part 2, dropout = 0.1 was the best overall regularization strategy on the selected Arch C base, reaching Val F1 = 0.9099.

- The best combined regularization settings did not outperform the best individual settings, showing that regularization effects were not simply additive.

- The results suggest that RGB is stronger in raw object-patch classification, while IR benefits substantially from carefully selected augmentation.

6.1 Suggestions for Report 2

- Use class-weighted loss or sampling to address the severe class imbalance, especially for truck, motor, and light.

- Use the best RGB regularization result and best IR augmentation result as baselines for later transfer learning, autoencoder, and final-performance experiments.

- Add confusion matrices and per-class analysis to identify which classes are most affected by modality and class imbalance.

- For final blind-test notebooks, provide a clean prediction pipeline that accepts an unlabeled patch folder and emits class predictions.

References

1. Frigui, H. (2026). CSE 547 - Deep Learning Algorithms and Methods. University of Louisville.

2. Teledyne FLIR. FLIR ADAS Dataset v2.

3. Paszke, A. et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library.

4. TorchVision documentation. Image transformations and pretrained model utilities.

Note: The code, trained checkpoints, generated manifests, and figures referenced in this report are available in the accompanying project directory and notebooks.